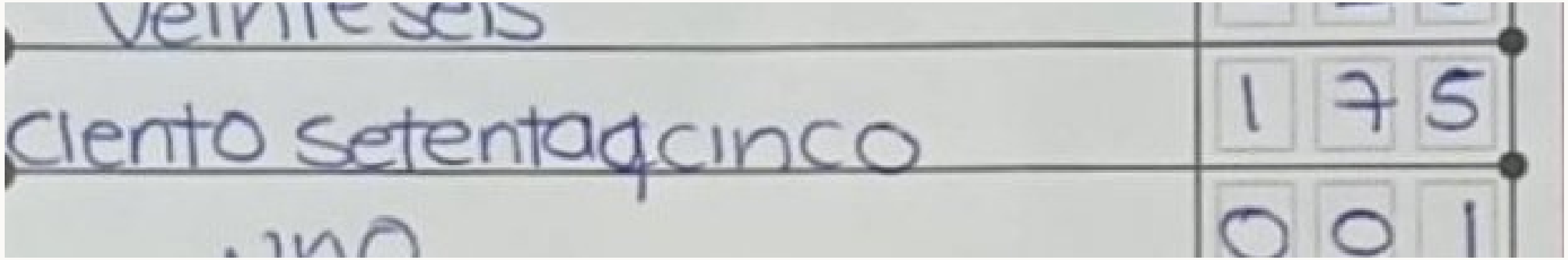
Two Hands: verifiable quality control for handwritten numbers

OCR reads the number. Two Hands checks whether the whole form agrees.

A second keyer that shows its proof. An open vision model, **Gemma on AMD**, re-reads every field of forms already keyed; the paper's arithmetic decides what is accepted; a person sees every disputed field with its photo.

Measured on a pre-registered random 300 of Mexico's 2024 handwritten tally sheets. An independent demonstration of clerical capture quality, not an official INE product.

One receipt



THE PAPER

175

words "ciento setenta y cinco" = one hundred seventy-five · boxes 1 7 5

THE FIRST CAPTURE KEYED

~~75~~

one of 52 fields where the capture does not match the paper

THE CERTIFIED COUNT

175

settled by comparison with the paper ("Cotejo")

Form 050815B0100, Option 07, from the pre-registered random 300 (masked crop). Sources: app/data/headline.json ("receipt"); INE PREP 2024 and Cómputos Distritales 2024, free-use terms. Each receipt is Merkle-anchored: "verify this receipt" in the app.

The problem, from INE's own files

6.5%
of the 52,167 polling stations that were not recounted: INE's two official captures of the same paper disagree on at least one vote field (3,382).

19.2%
of fully captured national forms fail their own row sum in the official capture (26,336 of 137,196).

software, not values
The mandated outside audit of each PREP, by rule, checks the software, not the keyed values against the paper.

A difference says two captures disagree, not which one is right; the photo does.

The preliminary count is preliminary by design; the certified count is the official result.

Sources: results/national_counts.md (PREP 2024 CSV; Cómputos Distritales 2024 CSV); audit scope: Reglamento de Elecciones art. 347.1 and INE's 2024 PREP audit report (UAM-I), PDF p. 7. Clerical only: no party, no totals, no outcome.

How it works: two hands and a referee

HAND 1

Gemma-4-31B-IT on AMD reads every field twice, the handwritten digits and the amount in words, plus each row's printed label.

HAND 2

A second, independent reader (Qwen3.8-27B on AMD) must agree field by field. It never overrides Gemma; it only withholds acceptance.

THE REFEREE

The paper's arithmetic: words = digits, rows sum to TOTAL, sections agree, every row label where it belongs.

The gate. Every identity holds: accepted. Any fails: the form is routed, with the failing field named. It never guesses a value.

A person sees the rest: the field, both reads and the crop. One value typed; the identities re-check at once. Names column masked before anything is stored or sent.

A form type is one schema file. Pipeline: locate the page and mask it (CPU) → one request per form on AMD Token Factory → gate (plain Python). docs/ARCHITECTURE.md.

Measured, not promised

52
fields where the official capture does not match the paper, each with its photo; **22** confirmed by the certified count, 30 judged from the photo by the AI adjudicator

ONE READER (PRIMARY)
18%
of forms accepted with no person: 2 wrong of 1,210 fields

TWO READERS (SECONDARY)
30%
0 wrong of 1,958 fields (lighter gate; full gate: 7%, 0 of 484)

CONFIRMATION, 100 FRESH FORMS
0 / 484
accepted fields wrong; the point estimate meets the pre-registered 0.5% bar (95% upper bound 0.79%)

Alone, one model mis-keys 1 field in 37. Behind the gate, 1 in 605 accepted fields is wrong.

Against the incumbent: 91.6% field match on Baja California Sur's local actas vs 62.84% for the Google Document AI reading the state bought (same metric). Recall 27 of 28 on certified-detectable errors (pre-registered).

Pre-registered random 300 (protocol ae103dd, Amendments 1-3, committed before any model read); Gemma-4-31B-IT read 289. The 30% tier is a secondary analysis, defined (Amendment 2) after the one-reader result was seen and before the second reader ran; it uses the lighter gate (no section checks). Wrong = against the official capture; every disagreement checked on the photo; agreeing fields not individually checked. Confirmation: gate v2, seed 20261017, results/confirm100.md (truth: AI adjudicator on published crops, labelled). app/data/headline.json; results/two_reader_300.md.

Institutes already tried to buy machine reading

- INE put MXN 9.5M in its 2024 draft budget for an automated first capture service (PTO numeral 30), a different step from my check, and bought none (last line). (p. 4)
- IPN quoted MXN 19.9M for the service. (p. 4)
- The acceptance bar was 80% agreement on words and digits. (p. 5)
- The best entrant reached 75%; INE declined it as untested in a real election, and its advisers (COTAPREP) recommended a pilot. (pp. 5, 10)

Two Hands is an automated capture with a referee. For 2027 I offer an independent check of a finished capture, which needs no rule change; the capture inside the loop is the pilot COTAPREP requested, offered as a pilot on 2024's own forms.

83.5%

On that metric as I read it, Gemma's existing reads of my pre-registered 300 agree with the official capture in words and digits on 83.5% of fields (5,508 of 6,600; failed reads and blank or illegible captures count against me).

INE's 2024 acceptance metric as I read it from the report, computed on my existing reads after they were made; the definition was committed before computation; not the same test set or conditions as INE's trial.

Beside it, the result that decides acceptance: 0 wrong of 1,958 accepted fields (two readers, the paper's sums as referee), judged against the official capture; every disagreement checked on the photo; fields where the model and the capture agree were not individually checked.

Source: INE, PREP PEF 2023-2024, "Captura Automática" report, CCOE 10th session 26-08-2024 (repositoriodocumental.ine.mx, ccoe-10se-26-08-2024-p17-1.pdf). Metric: results/ine_metric_DEFINITION.md (committed before computation), results/ine_metric.md. Buyer, 2027: the outside auditor or the contracted PREP operator, for an independent check of a finished capture; no rule change needed (docs/BUSINESS.md).

Business at a glance

Buyer	Mexico's preliminary counts (PREP): INE and the 32 state institutes, each with an outside auditor (all 26 named in 2024 were universities or research centres). At least 13 of the 17 outsourced 2024 state PREPs went to three firms.
Why now	6 June 2027: federal midterm and 32 concurrent local elections; each state picks its PREP operator by about 6 December 2026. INE budgeted MXN 9.5M for automated capture in 2024 and bought none (80% bar); its advisers asked for a pilot.
Product	An independent field-by-field check of a finished capture against the photo; no rule change needed. API, CLI and MCP server.
Price	**$1 per sheet audited (proposed); first step: a paid shadow-mode pilot** on one state's 2027 count. AMD compute: $0.00014 per sheet.
Reviewer work	About one photo crop per sheet (0.92 measured) instead of re-keying about 22 fields.
Capacity	One shared AMD key: Colima's 982 located sheets (of 1,017) in about 3.4 hours, Morelos's 7,802 in about a day.
Example	Morelos paid MXN 26.6M to run its 2024 PREP (7,802 sheets), including an MXN 1.5M audit of the software, not of the keyed values. My check of every sheet: about MXN 143k ($7,800), 0.5% of that.
Competition	Second keying by a person: at least 0.8% of fields in my random sample still do not match the paper. On Baja California Sur's 2024 local actas, Gemma on AMD matched the published PREP values on 91.58% of fields (n = 3,622 fields on a pre-registered random sample of 300 of the 2,020 actas; Wilson 95% 90.63-92.44%), on the per-field exact-match metric where the state's purchased Google Document AI reading scored 62.84% (14,501 of 23,077 fields on all 2,020 actas, against the manual capture rather than the published values).
Next	The same public data in at least 7 more countries (Indonesia, Colombia, Peru; Guatemala, Kenya, Argentina vote in 2027). Health reports: DHIS2 runs national reporting in 70+ countries. The check is the form's own arithmetic, so it isn't tied to elections.

Free sample deliverable: Colima, the whole state (1,017 in the PREP, 982 read, 35 not located, listed), report attached. Sources: docs/BUSINESS.md, "Sources for the Business at a glance table"; price: app/pricing.py (proposal); measured numbers recomputed by make eval.

Gemma on AMD

Gemma-4-31B-IT is in both zero-error two-reader pairs (0 wrong of 1,958 accepted fields with Qwen3.8-27B; 0 wrong of 924 accepted fields with Qwen3.8-Flash-Next); the only pair without Gemma made errors (2 of 814). Every form of the 300, on AMD GPUs via AMD Token Factory: one request per form (page + 6 zoomed tiles).

ALL 169,005 NATIONAL FORMS

~$24

of AMD compute ($0.000144 per form, measured)

SERVER TIME PER FORM

83 s

HTTP round trip on the shared endpoint; 87 s wall clock includes fixed pacing

TOKENS PER FORM

2,609 / 804

in / out; 289 of 289 replies end in stop

Open-model bake-off, same forms, prompt and gate: Gemma-4-31B-IT is the per-dollar leader and accepts the most forms; Qwen3.8-Flash-Next agrees most with the capture, at about fifteen times the price.

Open model on AMD Token Factory	Gemma-4-31B-IT	Qwen3.8-27B	Qwen3.8-Flash-Next
Forms accepted, no person	**55 of 300**	31 of 300	17 of 128
Cost per 1,000 forms	**$0.14**	$0.64	$2.11
Time per form (Gemma, Qwen-27B: wall clock; Flash-Next: server)	**87 s**	241 s	120 s

results/bakeoff_300.md (Amendment 6), results/gemma_vs_qwen.md, results/calls_gemma.jsonl; docs/GEMMA.md. Every measured read of the 300 is Gemma on AMD. Server time and cost: results/amd_benchmark.md.

Evolus: where the case lives

Workspace py_tr (event code AMDACT3).

- **Agent "Intake clerk"** on Gemma-4-31B-IT, AMD Token Factory: drafts each finding "for the auditor to confirm". It never decides.
- **Workflows:** "Audit finding" (the live audit opens a case per strong-evidence dispute, at most 60 a day), "Review hand-off", "Record filing".
- **AI Tool** `translate`: every review task in English and Spanish. **Scripts** re-check the paper's identities with no AI.

REQUIREMENT 2, CONFIRMED

"Token Factory works fine on Evolus and does count for the Track."

The Evolus team, Discord, 2026-10-16

The decision stays with the paper's arithmetic, never with a model; Evolus records each run as a workflow audit.

docs/EVOLUS.md; evolus/ (built and driven over the Evolus REST API, docs/evolus-openapi.json). Workspace and model endpoint stay online until Nov 15, 2026 for the Evolus review.

The next markets, not measured

Why elections. Most paper records are private, by law or by contract: patient charts, loan files, drug registers, so nobody can publish an error rate on them. Mexico's tally sheets are the clearest exception: the photo, the keyed value and the certified count are all public. That's why I measured there. **Elections are my proof set, not my only market.**

1. **Health facility reports (DHIS2):** WHO's Data Quality Review recounts paper source documents against what was keyed; DHIS2 runs in 70+ countries; monthly forms carry their own totals.
2. **US mortgage post-closing QC:** lenders re-check at least 10% of closed loans against source documents every month; 4.4 million closed-end originations in 2023.
3. **Clinical-trial source data verification:** it corrects 1.1% of site data while monitoring takes 15-30% of trial costs.
4. **Peru:** ONPE's 2026 rule keys every acta twice, one of them by a virtual assistant; not measured by me.

The referee is arithmetic, and arithmetic has no language. Measured on Spanish handwriting and English ledgers; the only language-specific check is words against digits, and the reader is a multilingual model.

Sources: WHO DQR Module 3 (2020); dhis2.org/about; Kenya MOH 711A; Fannie Mae Selling Guide D1-3-01, D1-3-02; CFPB HMDA 2023; Lindblad et al., Clinical Trials 2014; ONPE RJ 000093-2026 (El Peruano, 6 Jun 2026). Links: docs/BUSINESS.md.

Limits and the failure case

- **SOBRANTES (unused ballots)** is the one graded field in no identity printed on the form: every accepted-field error on the 300 was SOBRANTES. Gate v3 (pre-registered) accepts it only when two independent reads agree; otherwise a person sees it.
- **Digit insertion:** on 070212B0100 the model read 3114 where the paper says 314; the frozen gate accepted it, gate v2 did not.
- **Harmonised reads:** in one call the model can make words and digits agree (010469C0400: words and digits 239, paper 234). One read is not two independent reads; the fix is a second reader.
- **Failure case:** a rainfall ledger read as one image shifted a row one year; sums still held, 26 of 870 accepted cells wrong. Tiles and the row-label check removed it.
- **Recall** on the capture errors the certified count reveals, Colima (pre-registered, a person's blind check): 27 of 28 caught, 96% (Wilson 95% 82-99%). One state, certified-detectable errors only; not total recall.

Safeguard: accept only when every identity holds; everything else goes to a person with the field named. Fail closed when the page cannot be located (11 of 300).

app/data/headline.json ("harmonised"); results/confirm100.md; results/colima/recall.md (PROTOCOL.md Amendment 9); results/bakeoff_300.md ("Wrong accepted fields"); README "Limits and failure cases"; docs/EVALUATION.md.

Two Hands: verifiable quality control for handwritten numbers.

OCR reads the number. Two Hands checks whether the whole form agrees.

A second keyer that shows its proof: two independent readers on AMD, and the paper's own arithmetic as the referee. Every disputed field goes to a person with its photo.

Live demo: add at submission · **Video:** add at submission · **Repo:** add at submission

Reproduce every number offline: `make eval`. Gemma-4-31B-IT on AMD Token Factory · Evolus workspace py_tr. Vote rows shown as Option 01 to 15; no party is named or compared.